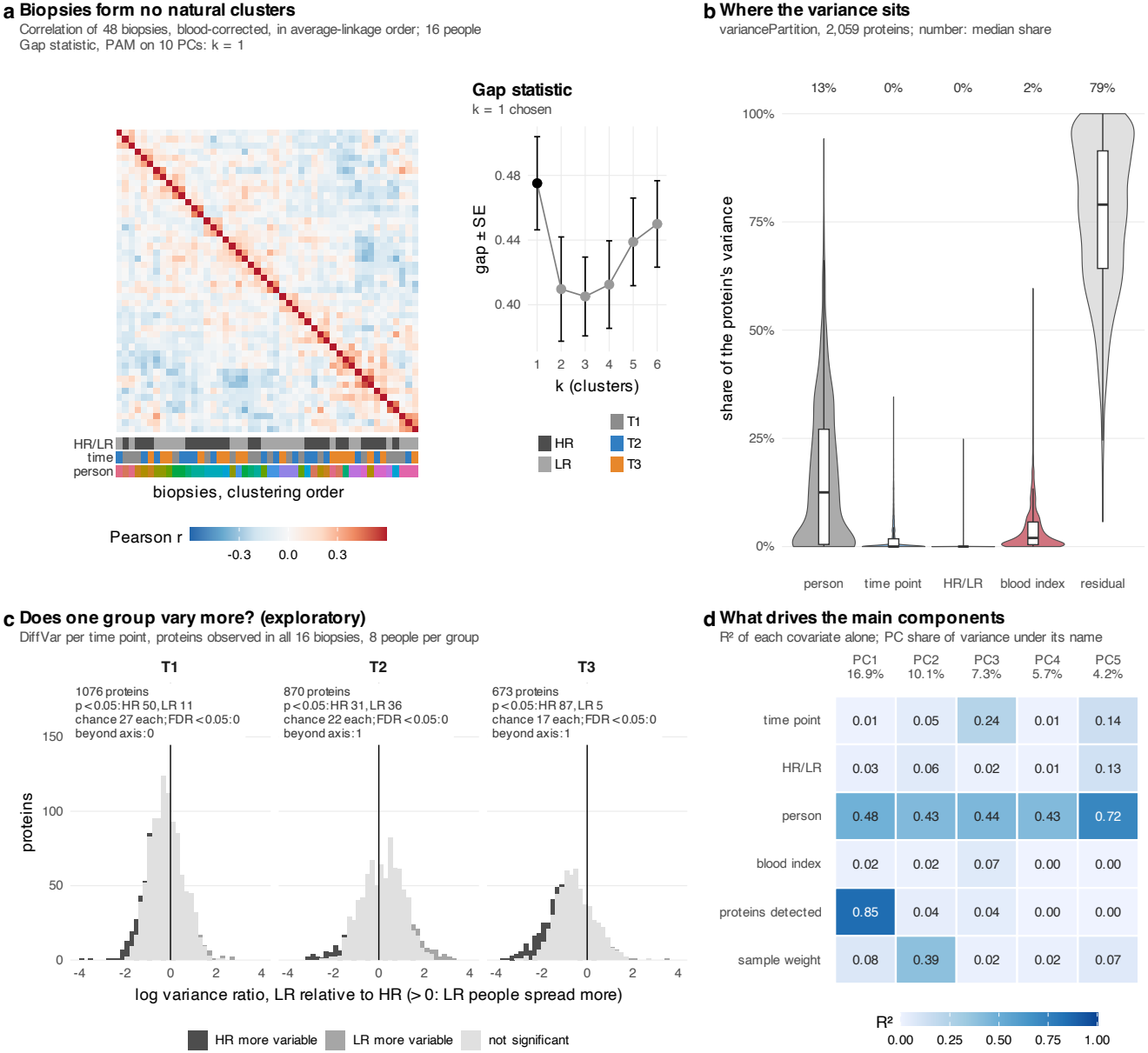

a Biopsies form no natural clusters
Correlation of 48 biopsies, blood-corrected, in average-linkage order; 16 people
Gap statistic, PAM on 10 PCs: k = 1
Gap statistic
k = 1 chosen
gap ± SE
0.48
0.44
0.40
1 2 3 4 5 6
k (clusters)
HR
LR
T1
T2
T3
HR/LR
time
person
biopsies, clustering order
Pearson r
-0.3 0.0 0.3
b Where the variance sits
variancePartition, 2,059 proteins; number: median share
13%
0%
0%
2%
79%
100%
75%
50%
25%
0%
share of the protein's variance
person
time point
HR/LR
blood index
residual
c Does one group vary more? (exploratory)
DiffVar per time point, proteins observed in all 16 biopsies, 8 people per group
T1
1076 proteins
p < 0.05: HR 50, LR 11
chance 27 each; FDR < 0.05: 0
beyond axis: 0
T2
870 proteins
p < 0.05: HR 31, LR 36
chance 22 each; FDR < 0.05: 0
beyond axis: 1
T3
673 proteins
p < 0.05: HR 87, LR 5
chance 17 each; FDR < 0.05: 0
beyond axis: 1
proteins
150
100
50
0
-4 -2 0 2 4
log variance ratio, LR relative to HR (> 0: LR people spread more)
HR more variable
LR more variable
not significant
d What drives the main components
R² of each covariate alone; PC share of variance under its name
PC1 16.9%
PC2 10.1%
PC3 7.3%
PC4 5.7%
PC5 4.2%
time point 0.01 0.05 0.24 0.01 0.14
HR/LR 0.03 0.06 0.02 0.01 0.13
person 0.48 0.43 0.44 0.43 0.72
blood index 0.02 0.02 0.07 0.00 0.00
proteins detected 0.85 0.04 0.04 0.00 0.00
sample weight 0.08 0.39 0.02 0.02 0.07
R²
0.00 0.25 0.50 0.75 1.00